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(1) Research Program/Project/Study Title: Critical Thinking Skills and Learning Engagement: Its Relationship towards the Academic Performance of Pre-Service Teachers	
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Program or Project has a component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation: DUMARAO SATELLITE COLLEGE									
Municipality/City:		Dumarao							
Province:		Capiz							
Region:		IV							
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
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		☐ Collaborative Research (external agency)
(7) Sector/Commodity		(8) Discipline/Field of Specialization EDUCATION-MATHEMATICS
(9A) Abstract Critical thinking is very essential for most college students. Acquiring a very good critical thinking skills will help students improve their thinking and enhance their engagement towards course of work, answering tests, completion of assignments, and projects in their courses which may lead in good academic standing. This study aimed to critical thinking skills and learning engagement: its relationship towards the academic performance of pre-service teachers. Specifically, it sought to answer the following: 1) What is the level of engagement, critical thinking skills and academic performance of pre-service teachers as a whole and when grouped according to profile?; 2) Are there significant differences on the level of learning engagement, critical thinking and academic performance when grouped according to profile; and, 3) Are their significant relationships between level of engagement, critical thinking skills and academic performance of pre-service teachers?. The study used descriptive-correlational research. The 56 pre-service teachers served as the respondents of the study. Cochran's formula was used in determining the sample size while stratified random sampling was employed in selecting the respondents. Mean, standard deviation, Kruskal-Wallis test, Mann-Whitney U test and Spearman rho were used in analyzing the data. Based on the findings of the study, the respondents have high level of engagement towards their studies, they also showcases high critical thinking skills which also resulting to their satisfactory academic performance. It was found out that male respondents however have dominant critical thinking skills and academic performance than females while it was noted that BSED major in Science revealed a significantly higher academic performance than BEED and BSED major in Social Studies. Further, pre-service teachers who showed greater commitment towards their studies and had acquired a significantly higher critical thinking skills were more likely to succeed academically. Intensification of learning engagement and strengthening of critical thinking skills were encourage to ascertain academic success among pre-service teachers.		
(9B) Keywords Learning engagement, critical thinking, academic performance and Pre-Service Teachers		
Introduction Critical Thinking is very essential for most college students. Critical Thinking skill will help students improve their thinking about their course of work, use thinking on tests, assignments, and projects in their courses, and have the strategic, analytical, problem solving, and decision-making skills they need when they opt to practice their chosen profession. Critical thinking is a common course in college and university settings today. Frequently taught as a way to "improve" thinking, the art of critical thinking involves an approach to thinking--more importantly to learning--that embraces changing how one thinks about thinking. Critical thinking incorporates how learners develop and apply thought to understand how thinking can be improved. Typically, a person is deemed a critical thinker to the extent that he or she regularly improves their thinking in an intentional manner. The basic idea under girding the study of critical thinking is simple to determine strengths and weaknesses in one's thinking to maintain the strengths and make improvements by targeting the weaknesses. Meanwhile, it was known for teachers that students who are actively engaged in learning especially during discussion possesses necessary critical thinking skills. Khlaif, et al. (2020) in their study students' engagement in face-to-face learning has a clear distinguished from students' engagement in online learning. Students are required to be involved and have a high level of engagement to obtain a flourishing experience in learning, which will impact academic outcomes. If students feel disengaged during learning, accomplishing everyday tasks will be challenging. As a result, the burden of a learning		

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environment will direct the student to skip the online program (Kahn et al., 2014). In addition, the level of student engagement could predict more emotional and psychological functioning.

As observed, being critical thinker give anyone an edge over those who have lack the skills. It is always observed that those who are capable of thinking outside the box are more than capable to accomplish the tasks assigned to them. It is expected of the aspiring professionals to have the necessary critical thinking ability and active engagement to succeed in the profession. Hence, this study.

Statement of the Problem

The study aimed to determine the relationship among critical thinking abilities, learning engagement and academic performance of college students. Specifically, it aimed to answer the following:

- 1) What is the level of learning engagement as a whole and when grouped according to age, sex and course?
- 2) What is the level of Critical thinking abilities as a whole and when grouped according to age, sex and course?
- 3) What is the academic performance of the respondents as a whole and when grouped according to age, sex and course?
- 4) Are there significant differences on the level of learning engagement when grouped according to age, sex and course?
- 5) Are there significant differences on the level of Critical thinking abilities when grouped according to age, sex and course?
- 6) Are there significant differences on the academic performance of the respondents when grouped according to age, sex and course?
- 7) Are there significant relationships between level of learning engagement, critical thinking abilities and academic performance?

Schematic Diagram

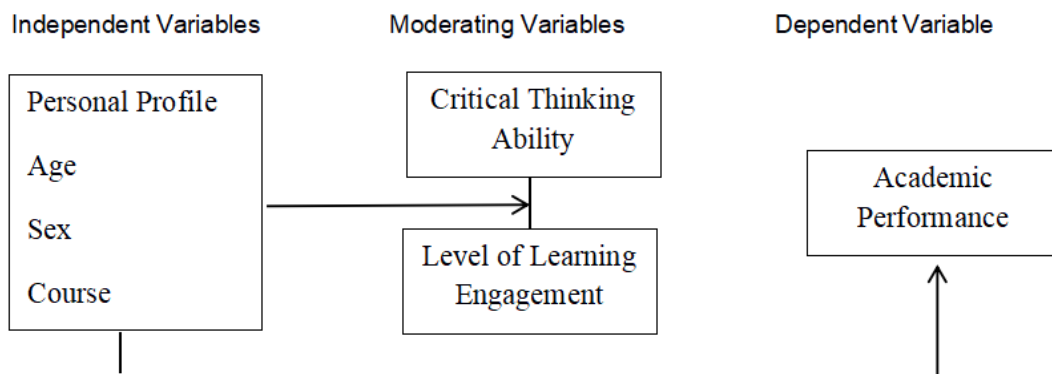


Figure 1. A Schematic diagram showing the relationship between independent, moderating and dependent variables.

Hypotheses

Based on the schematic diagram above, the following hypotheses were :

- 1). There are no significant differences on the level of learning engagement when grouped according to age, sex and course?
- 2). There are no significant differences on the level of Critical thinking abilities when grouped according to age, sex and course?
- 3). There are no significant differences on the academic performance of the respondents when grouped according to age, sex and course?

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4). There are no significant relationships between level of learning engagement, critical thinking abilities and academic performance?

Definition of Terms

Critical Thinking is defined as purposeful, self-calibrated judgment. This kind of judgment manifests itself in interpretation, analysis, evaluation, inference, and the explanation of the evidence, concept, method, standard, or context on which the judgment depends (Nair and Lynnette Leeseberg, 2013). In this study, it refers to the skills of students in truth-seeking, open-mindedness, analytic thinking, systematicity, inquisitiveness and cognitive maturity

Learning Engagement is defined as the range to which students engage in educational activities that correlate to high-quality learning outcomes (Krause and Coates (2008). In this study, it refers to the learning engagement of the pre-service teachers and were categorized into, excellent, high, moderate, low, very low.

Pre-Service Teacher

(11) Materials and Methods or Methodology

This descriptive-correlational research study utilized the 56 pre-service teachers enrolled at Capiz State University Dumarao Satellite College, Second Semester, School Year 2022-2023.

An adopted questionnaire on learning engagement and critical thinking skills was used to gather student responses, while the academic performance was taken from their previous GPA. The collected data were encoded in Microsoft Excel and processed in the SPSS software for analysis.

Mean, standard deviation, Kruskal-Wallis test, Mann-Whitney U test and Spearman rho were used to analyze the data.

(12) Results and Discussions

Level of Engagement

Table 1 shows the mean level of learning engagement among college students. The data revealed that the respondents in general had high level of learning engagement.

Age. It was found out that those who were 25 y/o and above got the highest mean level (M=3.9667; sd=0.15275), followed by 21-22 y/o (M=3.9227; sd= 0.49429) and the least mean was obtained by 23-24 y/o respondents (M=3.9138; sd=0.34583). All of the means however were interpreted as "high"

Sex. In terms of sex, female respondents got the higher mean level of engagement (M=3.9380; sd=0.36265) and males got a mean of 3.7660 and standard deviation of 0.99158, categorized as "high" respectively.

Course. As to course, it was found out that BSED major in Science respondents had highest mean level of learning engagement (M=3.8469; sd=0.42169); followed by BEED (M=3.7042; sd=0.27835) while the lowest mean was found in the BSED major in Social Studies (M=3.6750; sd= .37602) all categorized as "high".

Table 1. Mean level of learning engagement among college students

Profile	Mean	sd	Verbal interpretation
Entire Group	3.9227	.43990	High
Age			
21-22 y/o	3.9230	.49429	High
23-24 y/o	3.9138	.34583	High
25 y/o and above	3.9667	.15275	High

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Sex

Female	3.9380	.36265	High
Male	3.7660	.99158	High
Course			
BEED	3.7042	.27835	High
BSED SOC. STUD	3.6750	.37602	High
BSED SCIENCE	3.8469	.42169	High

Legend:

Very low	(1.0-1.80)
Low	(1.81-2.60)
Average	(2.61-3.40)
High	(3.41-4.20)
Excellent	(4.21-5.0)

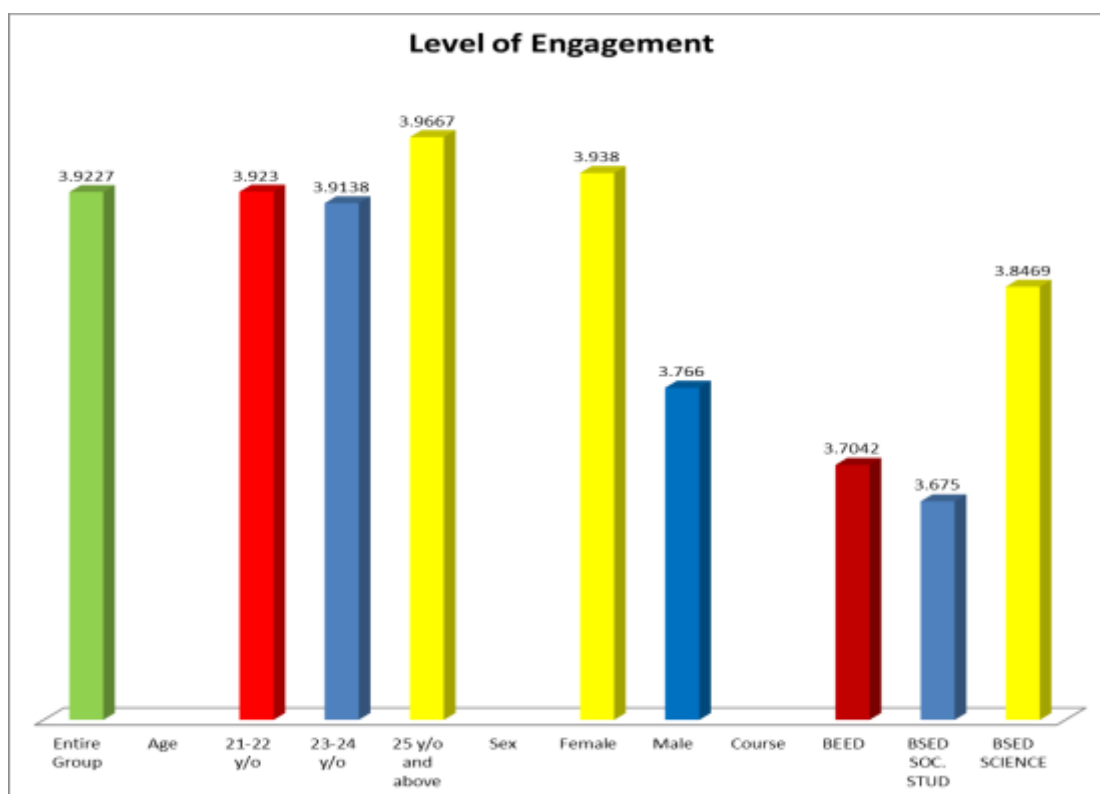


Figure 2. Bar graph showing the mean level of engagement of the pre-service teachers.

Level of Critical Thinking Skills

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Table 2 shows the mean level of critical thinking skills among college students. It was found out that the respondents in general had “high” level of critical thinking skills (M=3.7304; sd=0.37291).

Age. When the respondents were grouped according to age, 21-22 y/o got the highest mean level (M= 3.7635; sd=0.38562); followed by 25 y/o and above (M=3.7000; sd=0.30000) and the lowest mean level was found in 23-24 y/o respondents (M=3.6594; sd=0.36433). All of the mean scores were interpreted as “high”.

Sex. As to sex, males had higher mean score (M=4.1900; sd=0.22192) while female got the lower mean score (M=3.6853; sd=0.35472). Both scores however, were categorized as “high”.

Course. In terms of course, it was revealed that BSED major in Science got the highest mean score (M=4.0125; sd=0.57691), followed by BEED (M=3.9858; sd=0.23365) while the lowest mean score was found with BSED major in Social Studies (M=3.8443; sd=0.41729). All of the mean scores were categorized as “high”.

Table 2. Mean level of critical thinking skills among college students

Profile	Mean	sd	Verbal interpretation
Entire Group	3.7304	.37291	High
Age			
21-22 y/o	3.7635	.38562	High
23-24 y/o	3.6594	.36433	High
25 y/o and above	3.7000	.30000	High
Sex			
Female	3.6853	.35472	High
Male	4.1900	.22192	High
Course			
BEED	3.9858	.23365	High
BSED SOC. STUD	3.8443	.41729	High
BSED SCIENCE	4.0125	.57691	High

Legend:

Very low	(1.0-1.80)
Low	(1.81-2.60)
Average	(2.61-3.40)
High	(3.41-4.20)
Excellent	(4.21-5.0)

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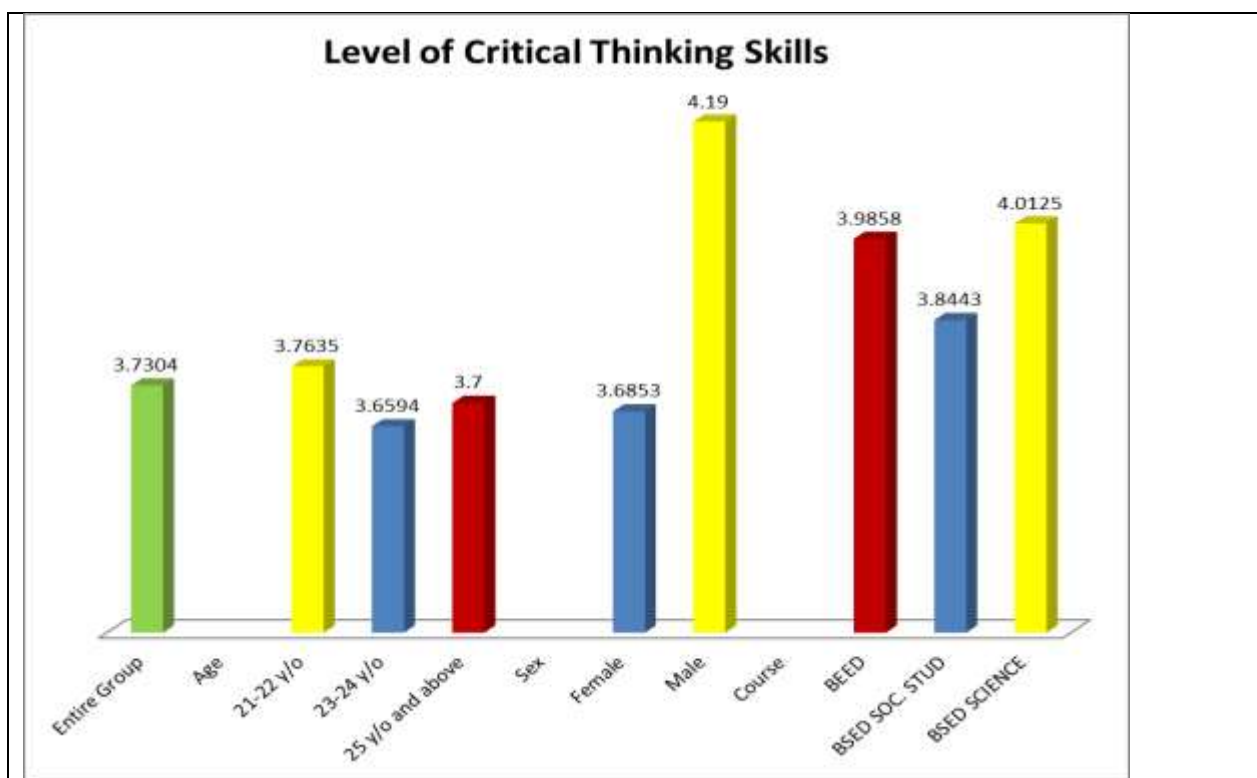


Figure 3. A bar graph showing the level of critical thinking skills of the pre-service teachers.

Level of Academic Performance

Table 3 shows the mean level of academic performance of the respondents. It was found out that the respondents had “satisfactory” academic performance ($M=2.0943$; $sd=0.23580$).

Age. In terms of age, 21-22 y/o got the highest mean score ($M=2.0789$; $sd=.24903$), followed by 23-24 y/o ($M=2.1413$; $sd=0.19809$) and the lowest was noted from 25 y/o and above respondents ($M=2.3333$; $sd= 0.30139$). All found to be “satisfactory”.

Sex. As to sex, males revealed a commanding very satisfactory academic performance ($M=1.7620$; $sd = 0.14377$) while females had “satisfactory” academic performance ($M=2.1269$; $sd= 0.21767$).

Course. When they were grouped according to course, BSED major in science revealed a “very satisfactory” academic standing ($M=1.9275$; $sd = 0.21663$), followed by BEED ($M=2.0517$; $sd =0.16579$) and the lowest GWA was shown by BSED social studies ($M=2.2079$; $sd= 0.21289$) which were interpreted as very satisfactory.

Table 3. Mean level of academic performance

Profile	Mean	sd	Verbal interpretation
Entire Group	2.0943	.23580	Satisfactory
Age			
21-22 y/o	2.0789	.24903	Satisfactory
23-24 y/o	2.1413	.19809	Satisfactory
25 y/o and above	2.3333	.30139	Satisfactory
Sex			
Female	2.1269	.21767	Satisfactory

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Male	1.7620	.14377	Very Satisfactory
Course			
BEED	2.0517	.16579	Satisfactory
BSED SOC. STUD	2.2079	.21289	Satisfactory
BSED SCIENCE	1.9275	.21663	Very Satisfactory

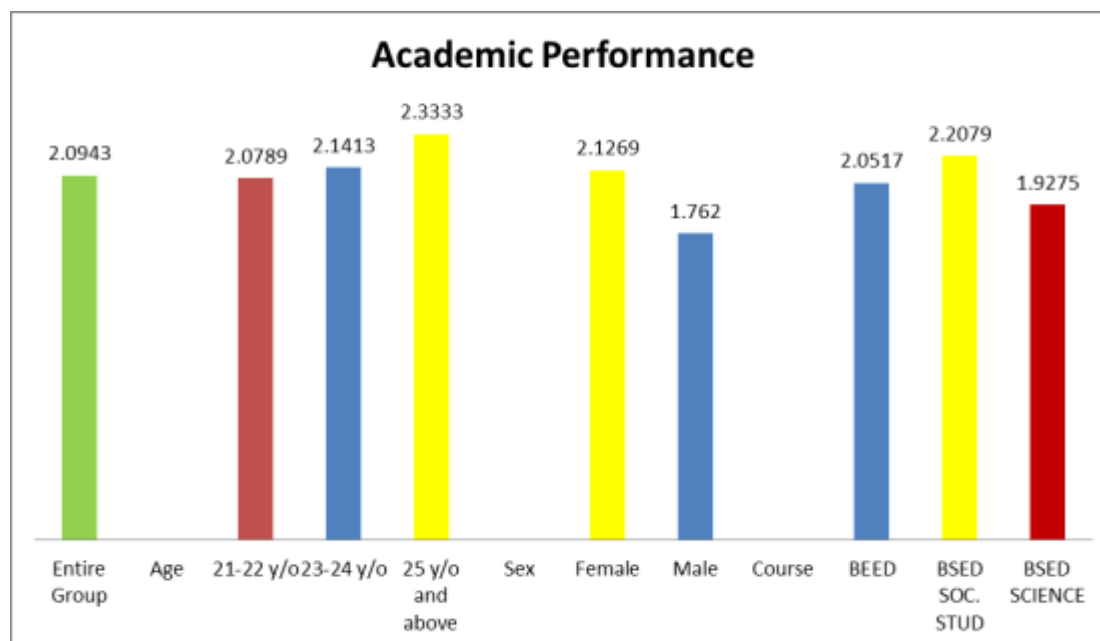


Figure 4. A bar graph showing the academic performance of the pre-service teachers.

Differences on the level of learning engagement when grouped according to profile.

The difference on the level of learning engagement among college students when grouped according to profile was shown in Table 4.

Age. Independent Samples Kruskal Wallis Test revealed a no significant difference on the level of engagement as to age ($\alpha=0.802$). This means that the level of engagement did not differ significantly with age. The null hypothesis which states that there is no significant difference on the level of engagement as to age was not rejected. Hence, regardless of the pre-service teachers age, they had high level of learning engagement towards their studies. This goes to show the commitment of students towards their studies regardless of their age. Thus, younger respondents have relative level of learning engagement with that of the older respondents.

Sex. Independent Samples Mann-Whitney U test revealed that there is no significant difference on the level of learning engagement when grouped according to sex ($\alpha=0.502$). Hence, the null hypothesis which states that there is no significant difference on the level of engagement as to sex was not rejected. This means that both sexes have relative level of learning engagement that, males have comparable commitment towards their learning so as with female pre-service teachers.

Course. Independent Samples Kruskal Wallis Test revealed a no significant difference on the level of engagement as to course ($\alpha=0.087$). This means that the level of engagement did not differ significantly with course. The null hypothesis which states that there is no significant difference on the level of engagement as to course was not rejected. Hence, regardless of the pre-service teachers course and specialization, they had high level of learning engagement towards their studies. This goes to show the commitment of students towards their studies regardless of their course. Thus, BEED respondents have relative level of learning engagement with that of the BSED respondents.

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Table 4. Differences on the level of learning engagement among college students when grouped according to profile.

Profile	Mean	test	sig	Decision
Age				
21-22 y/o	3.9230	Independent Samples Kruskal-Wallis Test	0.802 ^{ns}	Retain the null hypothesis
23-24 y/o	3.9138			
25 y/o and above	3.9667			
Sex				
Female	3.9380	Independent Samples Mann-Whitney U test	0.502 ^{ns}	Retain the null hypothesis
Male	3.7660			
Course				
BEED	3.7042	Independent Samples Kruskal-Wallis Test	0.087 ^{ns}	Retain the null hypothesis
BSED SOC. STUD	3.6750			
BSED SCIENCE	3.8469			

ns - not significant

Differences on the level of Critical thinking skills when grouped according to profile.

The difference on the level of Critical thinking skills when grouped according to profile was shown in Table 5.

Age. Independent Samples Kruskal Wallis Test revealed a no significant difference on the level of critical thinking skills as to age ($\alpha=0.699$). This means that the level of critical thinking skills did not differ significantly with age. The null hypothesis which states that there is no significant difference on the level of critical thinking skills as to age was not rejected. Hence, regardless of the pre-service teachers age, they had high level of critical thinking skills. Further, younger respondents have relative level of critical thinking skills with that of the older respondents.

Sex. Independent Samples Mann-Whitney U test revealed that there is significant difference on the level of learning critical thinking skills when grouped according to sex ($\alpha=0.003$). Hence, the null hypothesis which states that there is no significant difference on the level of critical thinking skills as to sex was rejected. It was found out that males have higher level of critical thinking skills than the female respondents. According to Sukarma et.al 2019, men are more likely to be able to think critically than women because critical thinking activities are logical, analytical, evaluative and reflective, but the importance of good emotional control in carrying out these activities carefully makes women also have the possibility to be superior to men in things critical thinking.

Course. Independent Samples Kruskal Wallis Test revealed a no significant difference on the level of critical thinking skills as to course ($\alpha=0.553$). This means that the level of critical thinking skills did not differ significantly with course. The null hypothesis which states that there is no significant difference on the level of critical thinking skills as to course was not rejected. Hence, regardless of the pre-service teachers course and specialization, they had comparable critical thinking skills. Further, BEED respondents have relative level of learning critical thinking skills with that of the BSED respondents.

Table 5. Differences on the level of critical thinking among college students when grouped according to profile.

Profile	Mean	test	sig	Decision
Age				
21-22 y/o	3.7635	Independent Samples Kruskal-Wallis Test	0.699 ^{ns}	Retain the null hypothesis
23-24 y/o	3.6594			
25 y/o and above	3.7000			

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Sex

Female	3.6853	Independent Samples Mann- Whitney U test	0.003**	Reject the null hypothesis
Male	4.1900			
Course				
BEED	3.9858	Independent Samples Kruskal- Wallis Test	0.553 ^{ns}	Retain the null hypothesis
BSED SOC. STUD	3.8443			
BSED SCIENCE	4.0125			

ns - not significant

*** - significant @ 1%*

Differences on the level of academic performance when grouped according to profile.

The difference on the level of academic performance when grouped according to profile was shown in Table 6.

Age. Independent Samples Kruskal Wallis Test revealed a no significant difference on the level of academic performance as to age ($\alpha=0.497$). This means that the level of academic performance did not differ significantly with age. The null hypothesis which states that there is no significant difference on the level of academic performance as to age was not rejected. Hence, regardless of the pre-service teachers age, they had high level of academic performance. Further, younger respondents have relative level of academic performance with that of the older respondents.

Sex. Independent Samples Mann-Whitney U test revealed that there is significant difference on the level of learning academic performance when grouped according to sex ($\alpha=0.002$). Hence, the null hypothesis which states that there is no significant difference on the level of academic performance as to sex was rejected. It was found out that males have higher level of academic performance than the female respondents. The result conforms to the study of Wrigley-Asante (2023) wherein gender differences in the academic performance at the senior high school level for all six subjects, with much of the differences occurring in Chemistry and Physics subjects, where males performed better than females. However, we found that the gender performance gap in favour of males switches in favour of females by the time they get to the tertiary level.

Course. Independent Samples Kruskal Wallis Test revealed a significant difference on the level of academic performance as to course ($\alpha=0.001^{**}$). This means that the level of academic performance did differs significantly with course. The null hypothesis which states that there is no significant difference on the level of academic performance as to course was rejected. It was found out that BSED science had significantly higher level of academic performance than BSED Social Studies but had a comparable academic standing with BEED while the latter had comparable General Weighted Average with BSED Social Studies.

Table 6. Differences on the level of academic performance among college students when grouped according to profile.

Profile	Mean	test	sig	Decision
Age				
21-22 y/o	2.0789	Independent Samples Kruskal-Wallis Test	0.497 ^{ns}	Retain the null hypothesis
23-24 y/o	2.1413			
25 y/o and above	2.3333			

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Sex

Female	2.1269	Independent Samples Mann-Whitney U test	0.002**	Reject the null hypothesis
Male	1.7620			

Course

BEED	2.0517 ^{ab}	Independent Samples Kruskal-Wallis Test	0.001**	Reject the null hypothesis
BSED SOC. STUD	2.2079 ^b			
BSED SCIENCE	1.9275 ^a			

ns - not significant

**-significant @ 1%

Relationships between level of learning engagement, critical thinking and academic performance

The relationships between level of learning engagement, critical thinking and academic performance was shown in Table 7.

Level of Learning Engagement and Academic Performance. It was found out that there is a strong positive correlation between the level of engagement and academic performance ($r = -0.367$; $\alpha = 0.005$). This means that the null hypothesis which states that there is no significant relationship between level of engagement and academic performance was rejected. Hence, the higher the level of engagement, the higher the academic performance. The result is parallel to the study of Abubakr et.al. 2018 which claimed that Students' engagement in school activities is a major factor in determining students' success in education, hence, the teachers need to undergo teacher training courses to obtain the necessary skill to dynamically engage students in classroom activities.

Level of Engagement and critical thinking skills. The test revealed that there is no significant relationship between level of engagement and critical thinking skills ($r = -0.014$; $\alpha = 0.916$). Hence the null hypothesis was rejected.

Level of critical thinking and academic performance. The test revealed that there is significant relationship between level of critical thinking and academic performance ($r = -0.547$; $\alpha = 0.000$). This positive relationship asserts that the higher the level of critical thinking skills, the higher the academic performance. Hence, critical thinking skills lead to academic success. The results conforms to the study of Q'Alessio et.al. 2019 which claimed that it is reasonable to advance that critical thinking has a positive impact on the average academic performance of MBA students. The analysis and interpretation capabilities are involved in the planning process, while the evaluation of arguments, inference, and deduction are important for decision-making.

Table 7. Relationship between level of learning engagement, critical thinking and academic performance among college students.

			Academic Performanc e	Level of Engagement	Critical Thinking Ability
Spearman's rho	Academic Performance	Correlation Coefficient	1.000	-.367**	-.547**
		Sig. (2-tailed)	.	.005	.000
		N	56	56	56

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Level of Engagement	Correlation Coefficient	-.367**	1.000	-.01
	Sig. (2-tailed)	.005	.	.916
	N	56	56	56
Critical Thinking Ability	Correlation Coefficient	-.547**	-.014	1.000
	Sig. (2-tailed)	.000	.916	.
	N	56	56	56

** . Correlation is significant at the 0.01 level (2-tailed).

(13) Conclusions

Out of the results and findings of the study the following conclusions were drawn.

- 1) . The respondents high level of learning engagement shows their high commitment towards their studies. Pre-service teacher had shown their best in order to deliver qualities and or characteristics expected of them in the practice of the teaching profession.
- 2) Critical Thinking skills are necessary in making decisions and solving problems. The “high” level critical thinking skills of the pre-service teacher indicate their capability in terms of truth-seeking, open-mindedness, analytic thinking, systematicity, inquisitiveness and cognitive maturity which were all the requisite skills necessary for the effective practice of the teaching profession.
- 3) The respondents revealed a “satisfactory” academic performance. This implies that the respondents do well in their studies. The general weighted average were all in compliance to the expectation of the Teacher Education department per retention policy.
- 4) The level of engagement did not differ by age, sex and course. Student level of commitment towards their studies were not affected of these variables.
- 5) Males have dominant critical thinking skills than female pre-service teachers. This indicate men's superiority in terms of critical thinking over their counterpart sex.
- 6) Coherently, males were more adept academically than females. This might be attributed to the fewer number of males enrolled plus the fact that despite their number, they were observed to be academically good. Further, BSSED major in Science had significantly higher academic performance than the rest of the course. This might be due to the fact those who are good in science were relatively well-versed in academic undertakings than any other course.

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7) Student level of engagement and critical thinking revealed to have strong positive correlations with academic performance. Academic success were attributable to these factors and hence the more motivated the students are to engage towards learning and the more critically minded they are, the more likely they are succeed academically.

Recommendations

Out of the conclusions given above, the following are suggested:

- 1) Students are encourage to intensify their commitment towards learning through active participation in both classroom and co-curricular activities.
- 2) Teachers may design creative teaching strategies to strengthen the critical thinking skills of students. Incorporation of thought-provoking questions in the discussion may help students think outside the box and develop their creative minds in solving problems.
- 3) Further studies related to this may be pursued particularly in determining the predictive attributes of student engagement and critical thinking skills towards performance in the licensure examination.

(14) Output

Baseline information for the study of the learning engagement, critical thinking and academic performance.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes, or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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